Sharp root log-concavity of alternating Motzkin sums

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Abstract

Let M_n be the Motzkin numbers and let $T_n = \sum_{j=0}^{n}(-1)^{n-j}M_j$. The sequence $T_n^{1/n}$ is strictly log-concave on its tail starting at $n = 5$. The inequality fails at center $n = 5$, so no earlier tail is log-concave. This proves Conjecture 1 of Zhao, which asks for log-concavity from index nine. The argument combines established ratio bounds with a size estimate. These estimates prove the infinite range; exact integer comparisons cover the remaining 25 centers. A Lean development proves the real-root inequality and its sharpness. An added source-definition module identifies the recursive implementation with the finite sum used in the conjecture.

Verification. The source theorem uses only `propext`, `Classical.choice`, and `Quot.sound`. The repository's verification record gives the exact builds, statement comparisons, independent kernel checks, and AI review status.

1 The problem and the result

A positive sequence (a_n) is log-concave if $a_n^2 \geq a_{n-1}a_{n+1}$ at each interior index. Strict log-concavity means that every such inequality is strict. For a tail starting at s, the interior indices are $n \geq s + 1$. This distinction between a tail index and a center index is needed below.

The Motzkin numbers satisfy the source recurrence and initial values:

$$M_0 = M_1 = 1, \qquad (n+3)M_{n+1} = (2n+3)M_n + 3nM_{n-1} \quad (n \geq 1). \tag{1}$$

These conditions determine the sequence uniquely over $\mathbb{Q}$ or $\mathbb{R}$. Define

$$T_n = \sum_{j=0}^{n}(-1)^{n-j}M_j. \tag{2}$$

The first values are

$$T_0, T_1, \ldots, T_9 = 1, 0, 2, 2, 7, 14, 37, 90, 233, 602.$$

Cancellation of consecutive sums gives

$$T_{n+1} = M_{n+1} - T_n. \tag{3}$$

Zhao studies this sequence in [5]. His Conjecture 1 asks whether $(T_n^{1/n})_{n\geq 9}$ is log-concave. The following theorem gives a sharper statement.

Theorem 1. *The values T_n are positive for $n \geq 2$. For every integer $n \geq 6$,*

$$T_n^{2(n^2-1)} > T_{n-1}^{n(n+1)}T_{n+1}^{n(n-1)}. \tag{4}$$

Thus $(T_n^{1/n})_{n\geq 5}$ is strictly log-concave. At center five the inequality is strictly reversed:

$$(T_5^{1/5})^2 < T_4^{1/4} T_6^{1/6}. \tag{5}$$

Consequently, five is the smallest starting index of a log-concave tail.

The ratio bounds used in the proof are established ingredients. The Motzkin bounds come from the interlacing method of Došlić and Veljan [2]. Zhao proves an interlacing estimate for the alternating sums [5, eqs. (7)–(8)]. Proofs of the needed estimates are included. General methods for root log-concavity also appear in Chen, Guo, and Wang [1] and in Hou and Li [3]. The contribution here is the stated alternating-sum result and its sharp threshold. No claim is made that the general method is new.

The proof has two parts. First, the proof bounds the logarithm of the ratio between the two sides of (4). The bound is positive for $n \geq 31$. Second, the proof compares exact integers for $6 \leq n \leq 30$. This covers every required center.

2 Ratio and size estimates

Put

$$x_n = \frac{M_{n+1}}{M_n}, \qquad b_n = \frac{6(n+1)}{2n+5}.$$

The recurrence shows by induction that $M_n > 0$ for every n.

Lemma 2. *For every integer $n \geq 1$,*

$$b_n \leq x_n \leq b_{n+1}. \tag{6}$$

Moreover, $0 < T_n \leq M_n$ for $n \geq 2$.

Proof. At $n = 1$, the values satisfy $x_1 = 2 = b_2$ and $b_1 \leq 2$. For $n \geq 2$, define

$$f_n(t) = \frac{2n+3}{n+3} + \frac{3n}{(n+3)t} \quad (t > 0).$$

The recurrence gives $x_n = f_n(x_{n-1})$. The function f_n is decreasing. Thus the induction hypothesis gives

$$f_n(b_n) \leq x_n \leq f_n(b_{n-1}).$$

Direct calculation yields

$$f_n(b_n) - b_n = \frac{3(n-2)}{2(n+1)(n+3)(2n+5)} \geq 0,$$

and

$$b_{n+1} - f_n(b_{n-1}) = \frac{9}{2(n+3)(2n+7)} > 0.$$

This proves (6). Since $b_n > 1$ for $n \geq 1$, the Motzkin numbers increase strictly from index one. Now $T_1 = 0$ and $T_2 = 2 = M_2$. If $0 \leq T_{n-1} \leq M_{n-1}$ and $n \geq 2$, then

$$M_n - M_{n-1} \leq T_n = M_n - T_{n-1} \leq M_n.$$

The lower bound is positive. This proves the remaining claim by induction. □

Lemma 3. *Let $y_n = T_{n+1}/T_n$ for $n \geq 2$. Then*

$$x_n \leq y_n \leq x_{n+1} \qquad (n \geq 9). \tag{7}$$

Proof. Equation (3) gives

$$y_{n+1} = x_{n+1} - 1 + \frac{x_{n+1}}{y_n}. \tag{8}$$

At $n = 9$, the values are

$$x_9 = \frac{2188}{835}, \quad y_9 = \frac{793}{301}, \quad x_{10} = \frac{2899}{1094}.$$

The two differences are 3567/251335 and 5057/329294, respectively. Both are positive.

Suppose (7) holds at $n \geq 9$. Its upper bound and (8) imply $y_{n+1} \geq x_{n+1}$. For the other bound, put

$$f(t) = \frac{2n+5}{n+4} + \frac{3(n+1)}{(n+4)t}, \quad c = \frac{2n+7}{n+5}, \quad d = \frac{3(n+2)}{n+5},$$

and define

$$H(t) = f(t) - 1 + \frac{f(t)}{t} - c - \frac{d}{f(t)} \quad (t > 0).$$

The functions $f(t)$, $f(t)/t$, and $-d/f(t)$ are decreasing. For the middle function, write it as $A/t + B/t^2$ with $A, B > 0$. Thus H is decreasing. The recurrence and the induction hypothesis give

$$\begin{aligned} y_{n+1} - x_{n+2} &\leq x_{n+1} - 1 + \frac{x_{n+1}}{x_n} - x_{n+2} \\ &= H(x_n) \leq H(b_n) \\ &= -\frac{3(4n^2 - 13n - 53)}{4(n+1)(n+4)(n+5)(2n+5)} < 0. \end{aligned}$$

The numerator is positive for $n \geq 9$. This proves the induction step. □

Lemma 4. *For every integer $n \geq 0$,*

$$M_n \leq \frac{3^n}{n+1}. \tag{9}$$

Proof. The cases $n = 0, 1$ hold directly. For $n \geq 1$, apply the recurrence and the bounds at n and $n-1$ to obtain

$$M_{n+1} \leq \frac{3^n(3n+4)}{(n+1)(n+3)} < \frac{3^{n+1}}{n+2}.$$

The strict comparison follows from $3(n+1)(n+3) - (3n+4)(n+2) = 2n+1 > 0$. □

3 The infinite range

For $n \geq 3$, put $q_n = T_n/T_{n-1}$ and define

$$D_n = n(n^2-1)\left(\frac{2\log T_n}{n} - \frac{\log T_{n-1}}{n-1} - \frac{\log T_{n+1}}{n+1}\right).$$

All values in the logarithms are positive by Lemma 2. The factor $n(n^2-1)$ is positive. Hence $D_n > 0$ is equivalent both to strict root log-concavity at center n and to (4). Substitution gives the useful identity

$$D_n = 2n \log q_n - n(n-1) \log \frac{q_{n+1}}{q_n} - 2 \log T_n. \tag{10}$$

For $n \geq 10$, Lemmas 2 and 3 give

$$q_n \geq b_{n-1} = \frac{3n}{n+3/2}, \qquad \frac{q_{n+1}}{q_n} \leq \frac{b_{n+2}}{b_{n-1}} = 1 + \frac{9}{n(2n+9)}.$$

Also, Lemmas 2 and 4 imply

$$\log T_n \leq n \log 3 - \log(n+1).$$

Use these estimates in (10). Since $\log(1+t) < t$ for $t > 0$,

$$\begin{aligned} D_n &\geq 2\log(n+1) - 2n \log\left(1 + \frac{3}{2n}\right) - n(n-1)\log\left(1 + \frac{9}{n(2n+9)}\right) \\ &> 2\log(n+1) - 3 - \frac{9(n-1)}{2n+9} =: F(n). \end{aligned} \tag{11}$$

For real $x \geq 10$,

$$F'(x) = \frac{8x^2 - 27x + 63}{(x+1)(2x+9)^2} > 0.$$

To give an exact positive value at the threshold, observe that

$$\log 2 = 2\int_0^{1/3} \frac{dt}{1-t^2} > 2\int_0^{1/3} (1+t^2)\, dt = \frac{56}{81}.$$

The strict inequality follows because $1/(1-t^2) > 1 + t^2$ for $0 < t < 1/3$. Consequently,

$$F(31) = 10\log 2 - 3 - \frac{270}{71} > \frac{560}{81} - 3 - \frac{270}{71} = \frac{637}{5751} > 0.$$

Monotonicity of F and (11) prove (4) for every integer $n \geq 31$.

4 The finite range and the sharp threshold

It remains to check the 25 centers $6 \leq n \leq 30$. Table 1 gives the values required for those comparisons, together with the value T_4 needed for sharpness. Every value follows from (1) and (3). Substitution in (4) gives a strict inequality at each of the 25 centers. These comparisons use integers only.

The following complete arithmetic procedure reproduces the finite comparisons. The division in the recurrence is checked to be exact.

```
M = [1, 1]
for n in range(1, 31):
    numerator = (2*n + 3)*M[n] + 3*n*M[n-1]
    assert numerator % (n + 3) == 0
    M.append(numerator // (n + 3))
T = [sum((-1)**(n-j)*M[j] for j in range(n+1))
     for n in range(32)]
for n in range(6, 31):
    assert T[n]**(2*(n*n-1)) > (
        T[n-1]**(n*(n+1))*T[n+1]**(n*(n-1)))
assert T[5]**48 < T[4]**30*T[6]**20
```

At center five, the reverse comparison follows from

$$37^{10} - 2^{24}7^9 = 4131563191399737 > 0.$$

Indeed, $T_4 = 7$, $T_5 = 14$, and $T_6 = 37$. Squaring the inequality and multiplying by 7^{30} gives $14^{48} < 7^{30}37^{20}$. It is equivalent to (5). Every tail starting at an index at most four contains center five. The infinite argument and the finite comparisons prove Theorem 1.

n	T_n	n	T_n
4	7	18	4805596
5	14	19	13393688
6	37	20	37458331
7	90	21	105089228
8	233	22	295673995
9	602	23	834086420
10	1586	24	2358641377
11	4212	25	6684761124
12	11299	26	18985057352
13	30536	27	54022715450
14	83098	28	154000562759
15	227474	29	439742222070
16	625993	30	1257643249141
17	1730786	31	3602118427250

Table 1: Exact alternating Motzkin sums.

5 Formal statements and reproducibility

The analytic Lean development defines rational Motzkin numbers through adjacent pairs. It defines the alternating sequence recursively through (3). A separate source-definition module proves uniqueness for (1) and identifies that recursion with (2). Thus its concluding theorem concerns the original finite sum. The recurrence assumptions in the general statement specify only the source sequence. They do not assume any estimate or part of the conclusion. The separate Challenge module imports only Mathlib. Its problem also requires existence of a sequence with the source recurrence. The Solution module proves this problem, including existence and the sharp threshold.

Table 2 gives the principal correspondence. Names are written relative to the namespace `MathAndTheory.Motzkin`.

Mathematical statement	Lean declaration
Source recurrence correspondence	`Source.motzkin_isMotzkin`; `Source.eq_motzkin_of_isMotzkin`
Source finite-sum correspondence	`Source.alternatingSum_eq_alternating`
Strict real-root inequality	`Source.finite_sum_roots_strict_log_concave`
Original tail-nine conjecture	`Source.zhao_conjecture_one`
Center-five failure	`Source.finite_sum_roots_sharp_failure`
Optimal tail start	`Source.finite_sum_tail_start_optimal`
Complete challenge	`Source.solution`

Table 2: Principal formal statements. The source-definition module passed its native Lean build and axiom audit.

The underlying modules prove the ratio estimates, positivity, size bound, and infinite range. They then use proof-producing exact arithmetic for the remaining indices. The root function is the actual real power $T_n^{1/n}$. The formal proof does not replace that function with an uninterpreted symbol or an assumed inequality.

An independent Python program also counts Motzkin paths and applies the finite alternating sum directly. It agrees with the published OEIS table through index 300 [4]. It checks the exact inequality at centers 6 through 100 and the reverse comparison at center five. These finite checks support the result. They do not prove the infinite range. The

Lean statement uses the recurrence characterization. It does not claim a formal proof of the lattice-path interpretation.

The proof and reproducibility files form one result family. The supporting ratio lemmas are not separate publication claims.

Authorship and AI use

Dev D. Goyal directed the project and is responsible for publication. The research proofs and original formalizations were produced with ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning, under his direction. AI assistance was also used for the subsequent mathematical audit, the source-definition proof, the independent computation, and this manuscript. These contributions do not constitute independent human expert review. The result has not been peer reviewed. No claim of first priority is made. The repository records the completed review and verification checks.

References

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